Velocity and position–time graphs

LEARNING TARGET I can calculate average velocity and interpret a position–time graph using slope.

A · KEY VOCABULARY

Average velocity	$\bar{v} = \Delta x \div \Delta t$. Carries direction; summarizes a whole interval.
Average speed	Total distance ÷ elapsed time. No direction.
Slope	On an x–t graph, slope = velocity (units m/s). Height = position, not velocity.
Instantaneous velocity	Velocity at one moment: the tangent slope on a curved x–t graph.

B · KEY EQUATION + UNITS

$$\bar{v} = \Delta x \div \Delta t$$

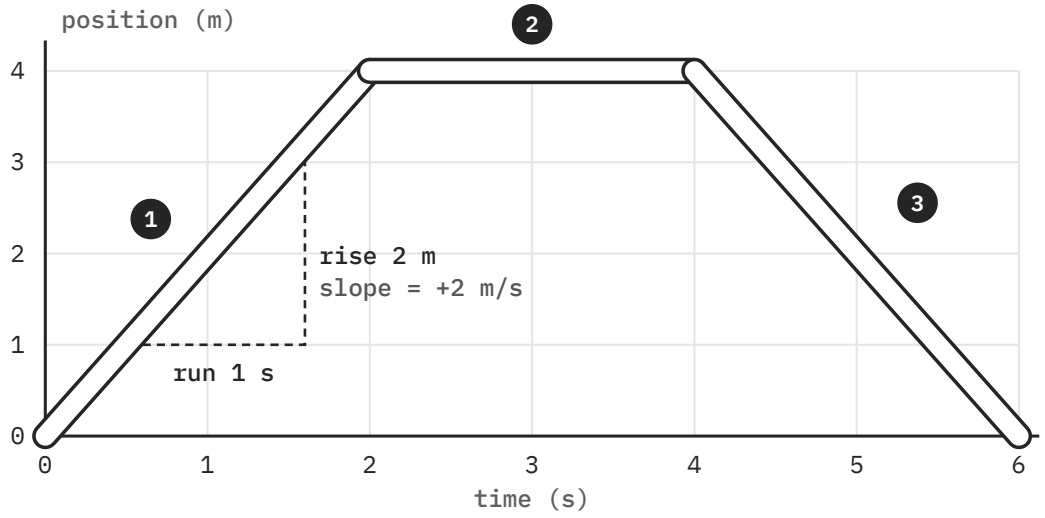
$$\Delta x = x_f - x_i$$

v velocity, meters per second (m/s)

speed total distance ÷ Δt

C · COLOR IT · READ A GRAPH AS A MOTION STORY

Height = position. Slope = velocity.



☐ **1 · PURPLE** Rising: moving +, $v > 0$

☐ **2 · PURPLE** Flat: stopped, $v = 0$

☐ **3 · PURPLE** Falling: moving –, $v < 0$

D · WORKED EXAMPLE

A straight x–t line passes through (2 s, 14 m) and (6 s, 6 m). Find the velocity, then predict x at 8 s.

GIVEN	EQUATION	WORK	ANSWER
(2 s, 14 m) (6 s, 6 m) predict at 8 s	$v = \Delta x \div \Delta t$ $x = x_i + v\,t$	$v = (6 - 14) \div (6 - 2)$ $x(8) = 6 + (-2)(2)$	**–2 m/s; 2 m** if the line continues

E · THE 2026-XJ THREAD

DAY 3 OF 22 · NEW TOOL: AVERAGE VELOCITY

Average velocity, $\Delta x \div \Delta t$ from two tracked positions, is the desk's first estimate of how fast XJ closes in. The slope of its x–t graph shows whether that rate holds.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else